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Paleomagnetism of the Lower Ordovician and Cambrian Sedimentary Rocks in the Section of the Narva River Right Bank: for the Construction of the Baltic Kinematic Model in the Early Paleozoic

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Abstract—The paleomagnetic study of the Lower Ordovician and Cambrian sedimentary rocks exposed on the Narva River's right bank revealed a multicomponent composition of natural remanent magnetization. Among four distinguished medium- and high-temperature magnetization components, the bipolar component, which carries the reversal test, is probably the primary component and reflects the geomagnetic field direction and variations during the Late Cambrian and Early Ordovician. The pole positions corresponding to this component have coordinates 22°N, 87°E (dp/dm = 5°/6°) for the Late Cambrian, and 18°N, 55°E (dp/dm = 5°/7°) for the Early Ordovician (Tremadocian and Arenigian). Together with the recently published paleomagnetic poles for the sections of the Early Ordovician in the Leningrad Region and the series of poles obtained when the Ordovician limestones were studied in Sweden, these poles form new key frameworks for the Upper Cambrian–Middle Ordovician segment of the apparent polar-wander path (APWP) for the Baltica. Based on these data, we propose a renewed version of the APWP segment: the model of the Baltica motion as its clockwise turn by 68° around the remote Euler pole. This motion around the great circle describes (with an error of A_{95} = 10°) both variations in the Baltic position from 500 to 456 Ma ago in paleolatitude and its turn relative to paleomeridians. According to the monopolar components of natural remanent magnetization detected in the Narva rocks, the South Pole positions are 2°S, 351°E (dp/dm = 5°/9°), 39°S, 327°E, (dp/dm = 4°/7°), and 42°S and 311°E (dp/dm = 9°/13°). It is assumed that these components reflect regional remagnetization events in the Silurian, Late Permian, and Triassic.

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INTRODUCTION

Present-day paleomagnetic data for the Lower Paleozoic of the Baltic–Scandinavian province indicate that it is still difficult to construct APWP for the interval from the Late Vendian including the Cambrian [Smethurst et al., 1998; Torsvik and Rehnstrom, 2001; Popov et al., 2002; 2005; Lewandowski and Abrahamsen, 2003; Nawrocki, 2004; Iosifidi et al., 2005]. However, a definite and correctly justified version of this APWP segment is a vitally necessary prerequisite for researchers to start discussing such problems as true polar wander (TPW), which was put forward in order to explain the high velocities of motion of lithospheric plates, following from paleotectonic reconstructions, which, in turn, were possibly responsible for the Cambrian revolution in the biosphere [Kirschvink et al., 1997]. This is equally important for the Early Paleozoic paleogeographic reconstructions describing the interrelations between Armorica, Avalonia, Baltica, and Gondwana [Bachtadse and Briden, 1990; Tait et al., 1995; Torsvik et al., 1996; Mac Niocaill et al., 1997; Torsvik and Rehnstrom, 2003]. However, all attempts to construct the APWP segment for the Cambrian and to join this period to the beginning of the Early Ordovician part of APWP were hindered by the low data quality and/or by a complete remagnetization of rocks [Torsvik and Rehnstrom, 2003; Nawrocki et al., 2004; Gurevich et al., 2005]. These problems are not unique for the Early Paleozoic but are also usual for the Vendian. Key poles, except the 555 Ma old pole for the Winter Coast [Popov et al., 2002; 2005] and the Vendian poles of the same age in Podolia and Volyn [Iosifidi et al., 2005; Nawrocki et al., 2004], are absent here. The present paper reports new paleomagnetic data on the series of the Early Ordovician and Cambrian sedimentary rocks exposed on the right bank of the Narva River below the Ivangorod fortress.

REGIONAL GEOLOGY AND SAMPLING

The Lower Paleozoic sedimentary rocks, exposed in the southeastern surrounding of the Baltic Shield, are represented by the 10–20 m sequence of the Cambrian and Tremadocian sandstone, sand, and clay and by the

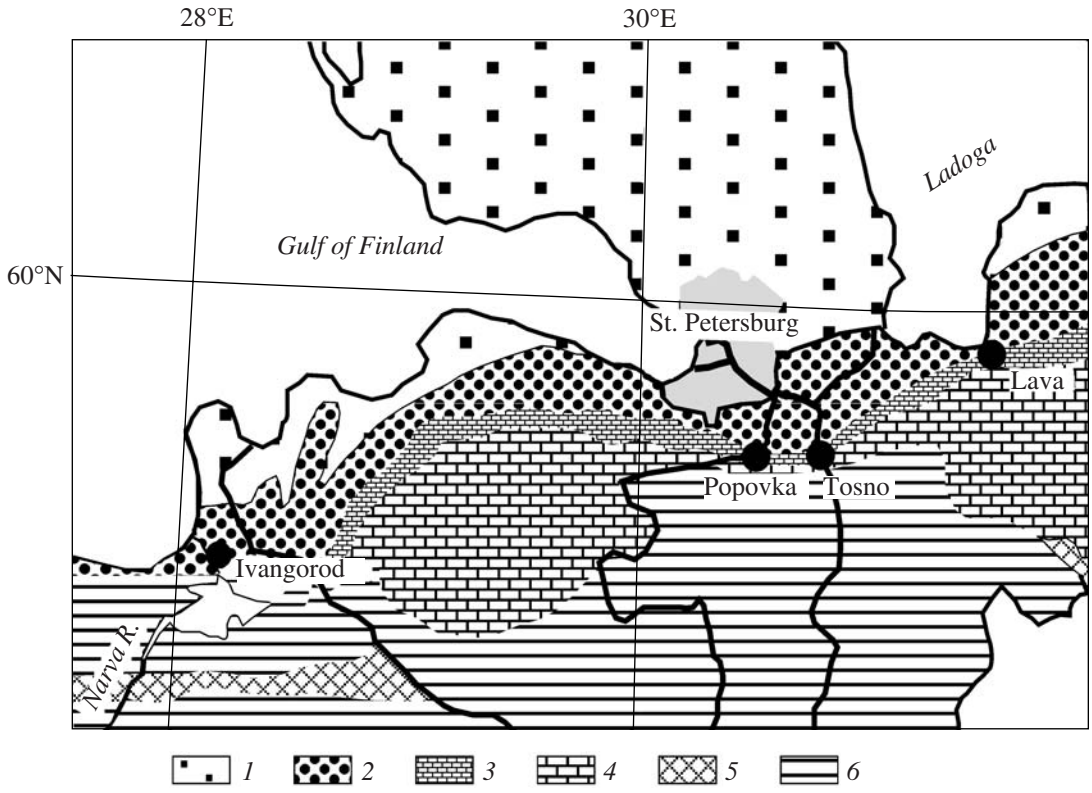


Fig. 1. Geological scheme of the working region. Rock age: (*1*) Precambrian; (*2*) Cambrian; (*3*) Early Ordovician; (*4*) Middle Ordovician; (*5*) Late Ordovician; (*6*) Devonian.

overlying 10–15 m sequence of the Arenigian and Llanvirnian limestones (Figs. 1, 2). These rocks are part of a thick (up to 2 km in depressions) Vendian–Devonian mantle, unconformably covering the Proterozoic rocks of the basement. The mantle rocks mainly occur subhorizontally and are only (in some places) disturbed by the Quaternary glacial dislocations or Devonian volcanic pipes. The Paleozoic section never submerged by more than several hundreds of meters [Selivanova and Kofman, 1971]. The Ordovician limestones are exposed at a low bench, which is traced along the southern coast of the Gulf of Finland, on the southern bank of the Neva River, and along the southern shore of Lake Ladoga (Fig. 1). This bench is the edge of the cover under which Tremadocian and Cambrian terrigenous deposits have been driven and maintained. The Ordovician limestones are extremely reach in fossils and fauna of trilobites, conodonts, and graptolites. Based on the sequence of the graptolite zones, the top section of the carbonate sequence was attributed to the Llanvirnian, whereas the Arenigian rocks undoubtedly compose the bottom section. The underlying terrigenous deposits include Tremadocian graptolites [Obut et al., 1991; Selivanova and Kofman, 1971]. In the eastern Baltic–Scandinavian province, the boundary between the Upper Cambrian and the Lower Ordovician is drawn within the Pakerort regional stage. The stage's lower boundary corresponds to the bottom of the Cordylodus andresi zone. In Scandinavia Cordylodus andresi appears in the top section of the Peltura zone, which is much lower than the Tremadocian's bottom section [Popov et al., 1997].

The sequence's bottom section represented (from top to bottom) by the Ladoga, Sablinka, Tiskre, and Lyukas groups, lies on the Nemakit–Daldyn rocks of the Siversk group (blue clay) [Popov et al., 1997]. Based on the complexes of conodonts, paraconodonts, and acritarchs as well as brachiopods, the Ladoga and Sablinka groups are compared with the Agnostis pisiformis (the Upper Cambrian) and Paradoxides paradoxisimus (the Middle Cambrian) Scandinavian zones, respectively. At the same time, the Tiskre and Lyukas groups represent the Mobergella–Schmidtiellus mickwitzi zone of the Atdaban age. The Middle and Upper Cambrian groups are most representative in the eastern part of the St. Petersburg region and were more eroded in the western direction; thus, the Tremadocian–Upper Cambrian beds unconformably overlie the Tiskre Lower Cambrian group in the Narva region [Popov et al., 1997] (see Fig. 2).

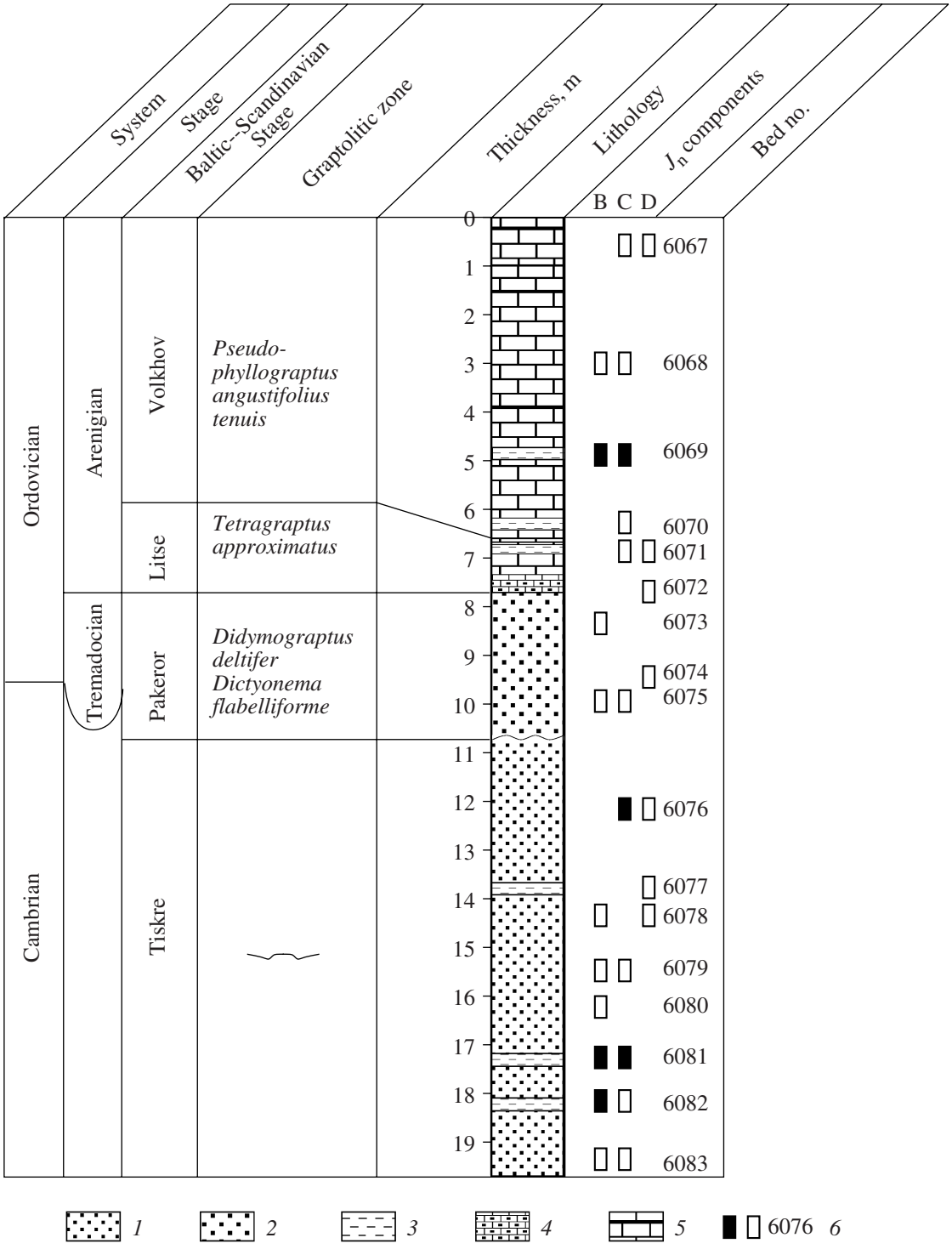


Fig. 2. Stratigraphic column for the Late Cambrian and Early Ordovician rocks along the Narva River: (*1*) sandstones; (*2*) obolus sandstones; (*3*) clays; (*4*) glauconitic sandstone; (*5*) limestones; (*6*) sampled beds and their numbers. The stratigraphic distribution of the B–D components; black (white) color correspond to normal (reverse) polarity.

Two sections (including the Arenigian, Tremadocian, and Cambrian groups) were sampled on the eastern bank of the Narva River below the Ivangorod fortress (Fig. 1). Forty six field samples (hand specimens), taken at 17 stratigraphic levels (sites), covered all stratigraphic units of the sequence in this region (Fig. 2).

LABORATORY STUDIES

Oriented field samples (hand specimens) were sawed into $2 \times 2 \times 2$ cm^3 cubic specimens. One hundred thirty six cubic samples obtained from 46 hand specimens taken at 17 stratigraphic levels, were subjected to stepwise thermal demagnetization (TH) and to demagnetization using an alternating magnetic field (AF). Thermal and AF demagnetization were performed using the TD48 and LDA-3 devices, respectively. S pinner magnetometers JR-4 and JR-5 were used to measure magnetization after each demagnetization step. Part of the collection was demagnetized at the paleomagnetic laboratory of the Munich University, where the Schonstedt devices and the 2G cryogenic magnetometer were used in a shielded room. The KLY-2 kappa bridge was used to measure magnetic susceptibility, including the control of the possible chemical changes in rocks during heating. The demagnetization results were analyzed using orthogonal projections [Zijderveld, 1967] and stereograms. The linear legs of demagnetization trajectories, determined at three (at the minimum) successive demagnetization steps, were identified approximately, and the trajectories of vector end points were subjected to a component analysis (PCA) during demagnetization [Kirschvink, 1980]. The average directions of the natural remanent magnetization components (J_n) and the related statistics were calculated according to the Fisher [Fisher, 1953] at the level of hand specimens and were subsequently averaged over the beds (sites). The software packages programmed by V.V. Popov (VNIGRI), Enkin [1994], and Torsvik and Smethurst [1999] were used to analyze and present the data. For characteristic samples, a thermomagnetic analysis was performed and the Curie temperatures and hysteretic properties were determined in order to identify rock magnetization carriers. These experiments were conducted in Munich, using the VFTB device. The technique of triaxial isothermal remanent magnetization (J_r) was applied [Lowrie, 1990]. According to this technique, magnetic fields of 1.4, 0.7, and 0.15 T were successively applied along the *X*, *Y*, and *Z* axes of a sample, respectively. These field values were determined based on the shape of the isothermal magnetization curves (Figs. 3a–3d) and taking into account the experience of the identification of remanent magnetization carriers in sedimentary rocks [Lowrie, 1990]. The following thermal demagnetization of the obtained complex J_r should have helped in identifying high-coercivity minerals (hematite or goethite) coexisting with softer ferromagnetic minerals (magnetite, pyrrhotite, or maghemite).

MAGNETIC PROPERTIES OF ROCKS

The initial values of the natural remanent magnetization (J_n) of the Narva rocks are rather different. However, these values form two definite groups comparable with lithology. The light gray and greenish limestones in the top section carry a very low J_n with an average value of 0.6 mA m^{-1}. On the contrary, reddish and purple sandstones and clays in the lower beds and a relatively small amount of purple limestones in the top section are characterized by high J_n (0.3–33 mA m^{-1}) with an average value of 3.2 mA m^{-1}. The hysteretic properties of these rocks are again shown in the same two groups. The [Lowrie, 1990] method in combination with the J_S-T experiments (Figs. 3e, 3f) indicated that magnetite is the main carrier of magnetization in carbonates (Figs. 3a, 3b, sample 6067-3), whereas hematite evidently predominates in the sandstone samples (Figs. 3c, 3d, sample 6080-3).

COMPONENT ANALYSIS OF ROCKS J_N

Up to four J_n components (from A to D) can be identified based on stepwise TH and AF demagnetization (Fig. 4). The process of AF demagnetization is effective when removing the J_n component of recent origin (component A), which is detected at the first demagnetization steps (Fig. 5a). In addition, AF demagnetization successfully detects the more stable J_n parts (components B–D) in the gray samples in the field varying from 20 to 80 mT. However, AF demagnetization is ineffective in samples of red rocks due to the presence of high-coercivity J_n carriers, whereas TH demagnetization was successfully used to separate components A–C in rock samples of any color. Therefore, TH demagnetization was selected as a standard procedure for the entire collection.

To determine the nature of component A, we held the samples during three years at different positions relative to the laboratory magnetic field before demagnetization. However, the directions detected as difference vectors in the process of thermal demagnetization from 150 to 250°C did not give a reasonable average direction, which leads to the conclusion that this component is of an exclusively viscous origin (Fig. 5b). Bipolar component B predominates in reddish and purple limestones (Fig. 4a) and sandstones (Fig. 4b); its blocking temperatures reach 685°C. The normal and reverse directions, oriented southwestward (SW) or northeastward (NE) down, were combined into subgroups B1 and B2, respectively, since bipolar component B does not withstand the reversal test (see below).

Component C is also bipolar, having southeastward (SE) downward and northwestward (NW) upward directions. Component C was detected in almost the entire section, with unblocking temperatures varying from 200 to 560°C in limestones (Figs. 4c, 4d) and reaching 670°C in sandstones (Figs. 4e–4g). To emphasize the difference between the C directions detected in the Ordovician and Cambrian rocks, we marked these directions by C1 (Ordovician) and C2 (Cambrian). Component D, directed toward the southwest (SW) downward, is detected in the intermediate temperature intervals of thermal demagnetization; this component is present in limestones in the top section and in sandstones in the bottom section's upper part.

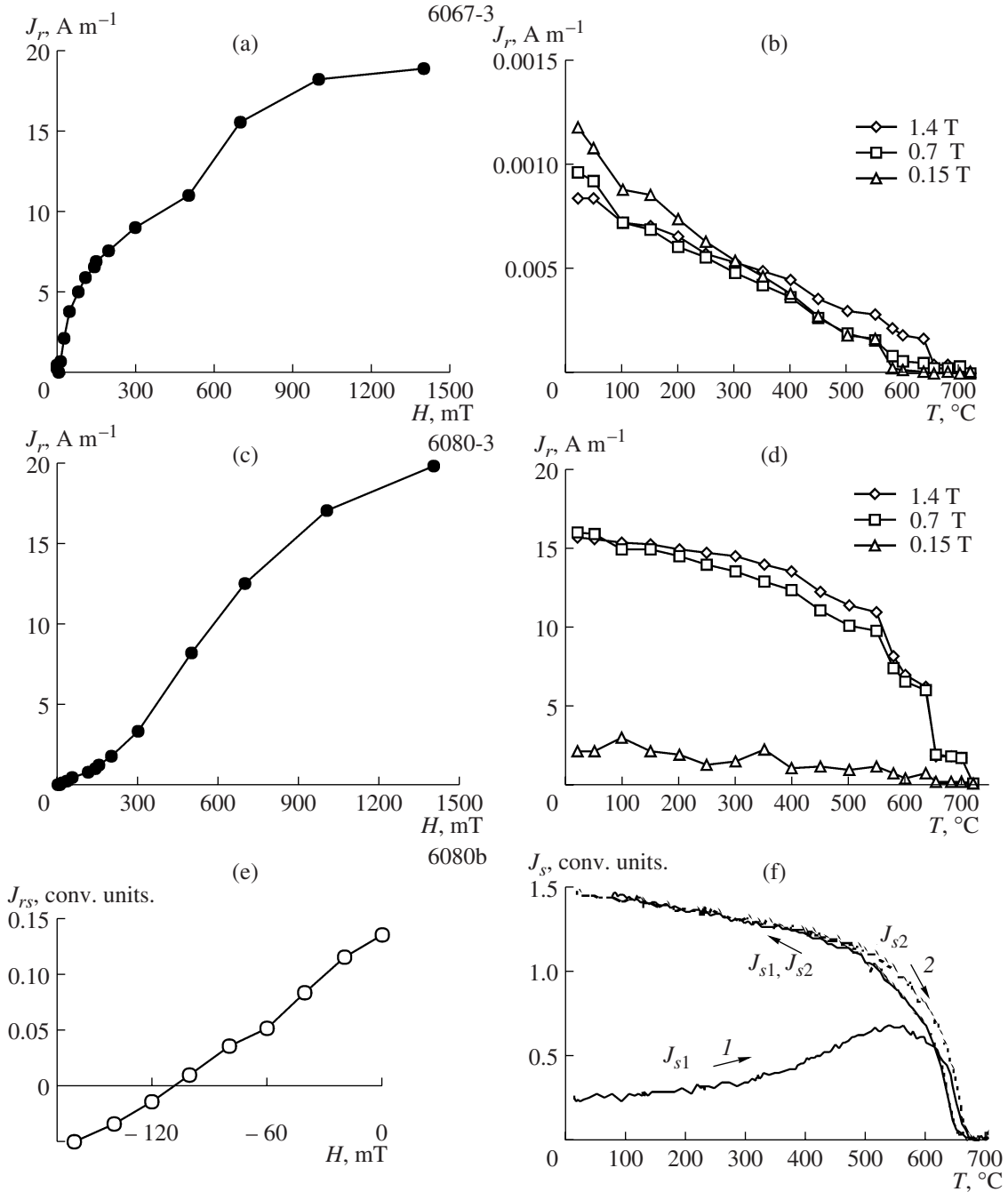


Fig. 3. Determination of ferromagnetic minerals according to [Lowrie, 1990] and using the VFTB device: (a) and (b) the curves of isothermal remanent magnetization (IRM); (c) and (d) the curves of thermal demagnetization of hard and soft IRM components formed during magnetization in the fields of 1.5, 0.7, and 0.15 T along the *X*, *Y*, and *Z* axes, respectively; H_{cs} is the destroying field; J_{s1} and J_{s2} are the curves of the first and second heating (cooling), respectively.

Component A and at least one of the B–D components are present in each sample (Fig. 4). However, the spectra of unblocking temperatures of these components are highly variable and depend on the rock type. While the B2 component in limestones has larger values of unblocking temperatures than components B1, C1, and D, component C2 in sandstones was detected in the temperature interval similar to such a interval for B2 (Table 1, Fig. 4).

The distributions of directions of the B–D components are shown in Figs. 5c–5f. The average directions of these components and the statistical parameters for each bed in the stratigraphic succession are given in Table 1.

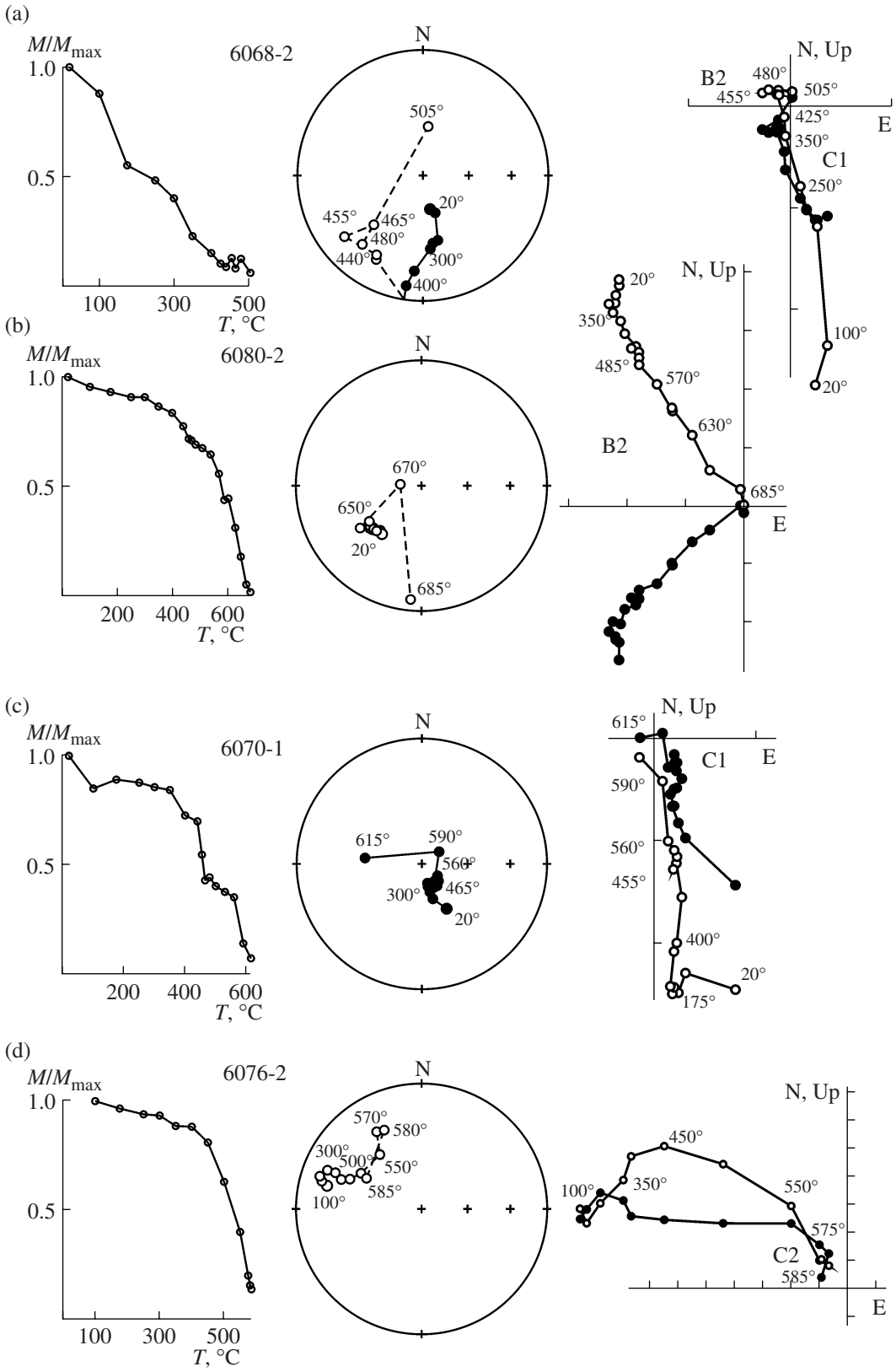


Fig. 4. Examples of the Zijderveld diagrams of the Narva rock sample demagnetization by heating: (a), (c), (f), and (h) the Early Ordovician; (b) and (d)–(f) the Cambrian rocks (open circles are the vector projections in the vertical conditionally meridional plane; filled circles are the vector projections in the horizontal plane).

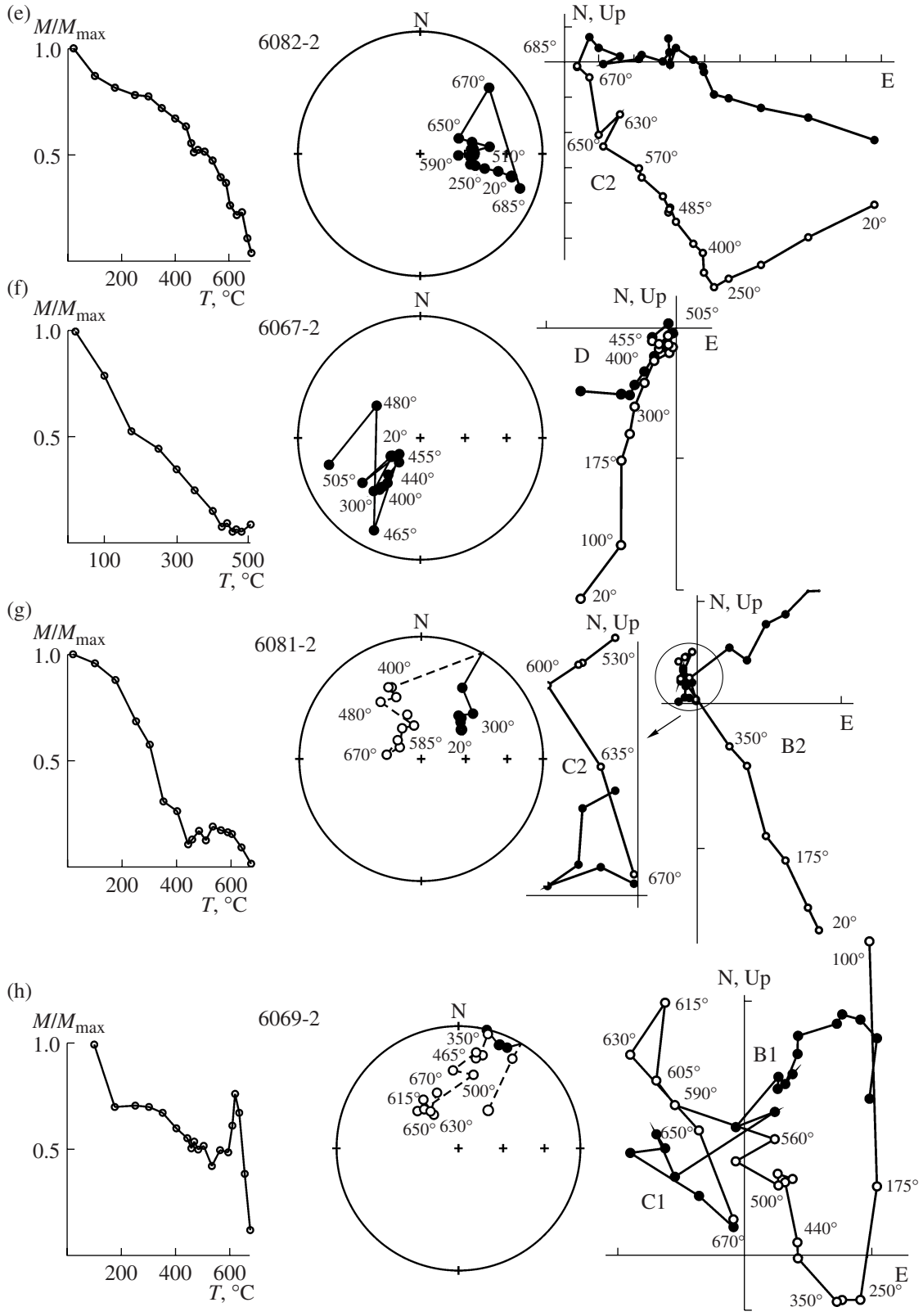

(e)
M/M_max
1.0
0.5
200 400 600
T, °C
6082-2
N
670°
650°
590°
510°
250°
20°
685°
N, Up
685°
670°
E
630°
650°
570°
C2
485°
20°
400°
250°
(f)
M/M_max
1.0
0.5
100 300 500
T, °C
6067-2
N
480°
20°
455°
505°
440°
300°
400°
465°
N, Up
505°
455°
E
D
400°
300°
175°
100°
20°
(g)
M/M_max
1.0
0.5
200 400 600
T, °C
6081-2
N
400°
480°
300°
670°
585°
20°
N, Up
600°
530°
E
350°
B2
C2
635°
175°
670°
20°
100°
(h)
M/M_max
1.0
0.5
200 400 600
T, °C
6069-2
N
350°
465°
670°
500°
615°
650°
630°
N, Up
615°
630°
B1
605°
590°
650°
560°
175°
C1
500°
440°
670°
E
350°
250°

Fig. 4. Contd.

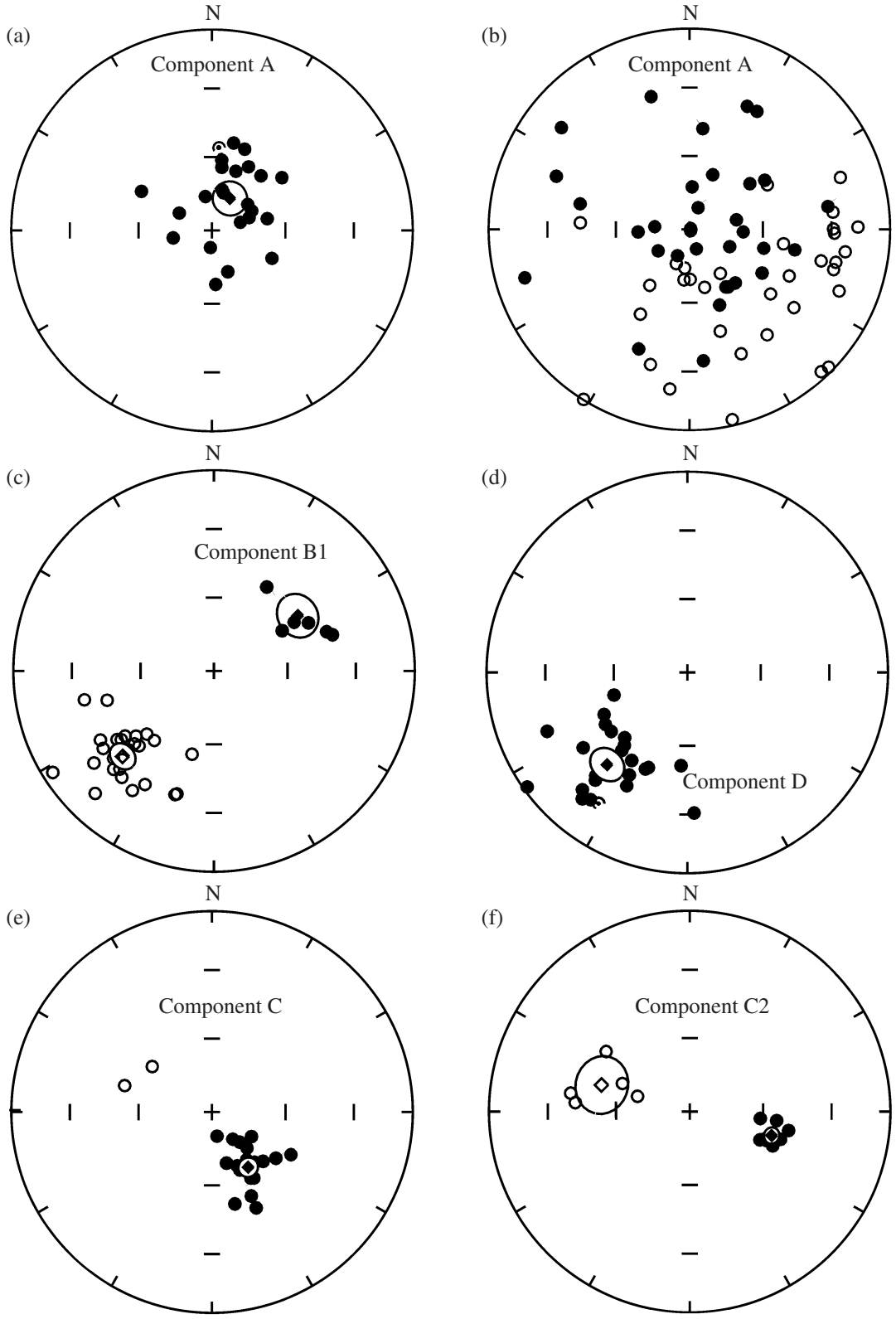


Fig. 5. Distribution of the directions of the natural remanent magnetization components of the Cambrian and Early Ordovician sedimentary rocks along the Narva River. The directions averaged for hand specimens (presented in Table 1) are shown for each J_n component; open (filled) circles are the vector projections onto the upper (lower) hemisphere. The remaining explanations are given in the text.

POSSIBLE AGE OF J_n COMPONENTS

Component A is undoubtedly young since its average direction in the AF demagnetization tests is identical to the direction of the local geomagnetic field because of the viscous origin of this component estimated during the extract tests (Figs. 5a, 5b). Components B1 and B2 are characterized by relatively high unblocking temperatures and are often the only available components in red rocks (Fig. 4b). The petromagnetic data indicate that the coercitivity of the magnetic material is very high (Figs. 3e, 3f); consequently, fine hematite carries the magnetization of this material. Field observations indicated that this fine hematite forms red spots and irregular streaks in beds and red pigment in fractures in limestones. Therefore, we can assume that B-magnetization is related to the surface or (and) subsurface processes of weathering under arid conditions that existed during the Permian–Triassic in this region. The negative result of the reversal test [McFadden and McElhinny, 1990], when an angle of 13° between the axes of the B1 and B2 components exceeds the critical angle (11°), indicates that a relatively long interval exists between two demagnetization events (Table 2). The ages of the B1 and B2 components is determined based on the position of the corresponding poles, which are located close to the Permian–Triassic APWP segment for Baltica [Smethurst et al., 1998].

Component C2 is bipolar and carries the reversal test (class C, an angle of 3° between the average directions of the N and R axes at a critical angle of 14°; see Table 2). Component C1 was detected only in the Lower Ordovician carbonate rocks, where diagenetic changes are minimal, and is therefore interpreted as a primary or early diagenetic component. Consequently, the Early Ordovician age of the C1 component is most probable. The fact that the obtained pole positions for the C1 and C2 components have no similarity among the younger poles makes this interpretation even more convincing. However, the age of the C2 component in the Early Cambrian rocks is less evident. The closeness of the directions of the C2 component, distinguished in the Tiskre group of the Lower Cambrian, to such directions for the Pakerort Tremadocian Formation of the Upper Cambrian (see Table 1, bed 6075) indicates that the Tiskre group was probably remagnetized in the Late Cambrian. This interpretation seems to be logical if we take into account the deltaic character of the Tiskre rocks, which were close to the ground surface or were exposed during a considerable period up to the Tremadocian [Popov et al., 1997]. Substantial differences between the C1 and C2 directions (Table 2) indicate that the assumption of the pre-Ordovician age of the C2 component is valid.

The age of component D, which is present in seven beds of the section (Fig. 5, Table 1), is more uncertain. The absence of a statistically significant difference between the D directions in the top and bottom sections (Tables 1, 2) can indicate that the D component is secondary (Table 2). This assumption is supported by the fact that paleopole D is close to the Silurian–Early Devonian APWP segment for the Baltic region. The Old Red continent, which included the region studied, existed at that time. Therefore, we can assume that component D, like component B, was also acquired under dry, arid conditions. However, we cannot exclude that this component (as an intermediate one) can have no geochronological sense, having originated as an artifact from a wide overlapping of the unblocking spectra of components B and C.

Figure 2 indicates the stratigraphic positions of the B–D components of natural remanent magnetization in the summary section (the B–D denotations are given on the right). The distribution of component C over the section indicates that reverse polarity predominates in the Arenigian and Tremadocian sections except in the two samples from bed 6069 in the middle section, which indicated normal polarity (see also Table 1). This pattern is evidently in rather good agreement with the magnetostratigraphic scheme for the Ordovician of the Baltica proposed in [Trench et al., 1991], according to which the interval of reverse polarity extends from the top of the Tremadocian to the middle of the Llanvirnian. The recent data on the Ordovician sections of Siberia and Baltica confirmed that a long interval of reverse polarity existed in the Ordovician [Torsvik and Trench, 1991a; Smethurst et al., 1998; Gallet and Pavlov, 1996; Rodionov et al., 2001]. Consequently, the magnetostratigraphic test supports our arguments for the primary origin of component C1.

The magnetostratigraphic succession established in the Lower Tremadocian–Cambrian section is more complex, and it is more difficult to compare this succession with the magnetostratigraphic data from other regions. The alternation of the zones of normal and reverse polarity in the sections of the rocks of this age in Siberia was determined repeatedly [Rodionov and Osipova, 1985; Khramov, 1987; Pisarevsky et al., 1998; Pavlov et al., 2000]. Based only on the predominance of reverse polarity in the Atdabanian rocks in northern Siberia [Pisarevsky et al., 1998], we would interpret component C as a reflection of the geomagnetic field polarity reversals registered in the Tiskre group, which is supposedly of the Atdabanian age [Popov et al., 1997]. Accepting our assumption of the Late Cambrian age of component C in the Tiskre rocks, we obtain again that reverse polarity predominates in the Narva and Siberian sections. If component C2 in the Lower Cambrian Narva rocks is secondary (Late Cambrian), we can assume that components B1 or B2 are primary. Indeed, it was previously assumed that the paleomagnetic directions close to the directions of the B1 component are primary and correspond to normal polarity for the Baltic Cambrian [Torsvik et al., 1996; Smethurst et al., 1998; Torsvik and Rehnstrom, 2001; 2003]. However, following the polarity selected by these researchers, we would have obtained that normal polarity predominates for component B in the Cambrian part

Table 1. Paleomagnetic directions and poles of the Cambrian and Early Ordovician Narva sedimentary rocks

	Stratigraphic level	B1 and B2 components							D component						C1 and C2 components						
			M/*N*/*n*	*T*, °C	D°	I°	K	α_{95}°	*M*/*N*/*n*	*T*, °C	D°	I°	K	α_{95}°		*M*/*N*/*n*	*T*, °C	D°	I°	K	α_{95}°
Arenigian	6067	–	–	–	–	–	–	–	3/8	250–450	220	37	39	20	C1	3/4	350–510	151	58	63	16
	6068	B2	3/7	400–500	203	–40	27	24	–	–	–	–	–	–	C1	5/8	300–440	158	60	36	13
	6069	B1	4/11	250–590	54	46	22	20	–	–	–	–	–	–	C1	2/2	600–670	296	–57		
	6070	–	–	–	–	–	–	–	–	–	–	–	–	–	C1	5/9	350–580	143	68	55	10
	6071	–	–	–	–	–	–	–	4/7	250–430	214	55	21	20	C1	6/12	350–590	145	64	123	6
	6072	–	–	–	–	–	–	–	5/13	450–600	226	31	30	14	–	–	–	–	–	–	–
Tremadocian	6073	B2	4/8	300–600	225	–34	93	10	–	–	–	–	–	–	–	–	–	–	–	–	–
	6074	–	–	–	–	–	–	–	5/8	450–600	228	40	16	20	–	–	–	–	–	–	–
	6075	B2	4/5	250–580	225	–31	18	22	–	–	–	–	–	–	C1	3/5	300–530	131	63	58	16
	average	B2	3*/11/20	–	218	–35	55	11	4*/17/36		223	41	51	10	C1R	5*/22/38	–	146	63	185	5
		B2	3/11*/20	–	219	–35	26	9	4/17*/36		223	41	20	8	C1N,R	6*/24/40	–	140	62	94	6
		B1	1/4*/11	–	54	46	22	20	–	–	–	–	–	–	C1N,R	6*/24/40	–	144	63	50	4
Cambrian	6076	–	–	–	–	–	–	–	2/4	400–530	203	44	–	–	C2	3/3	550–670	285	–44	41	20
	6077	–	–	–	–	–	–	–	2/2	440–570	229	26	–	–	–	–	–	–	–	–	–
	6078	B2	5/13	450–680	239	–40	32	14	4/9	250–480	221	31	14	26	–	–	–	–	–	–	–
	6079	B2	4/11	450–670	226	–40	30	17	–	–	–	–	–	–	C2	1/1	300–650	112	59	–	–
	6080	B2	4/14	350–670	229	–43	381	5	–	–	–	–	–	–	–	–	–	–	–	–	–
	6081	B1	1/2	300–590	59	58	–	–	–	–	–	–	–	–	C2	2/2	600–670	289	–64	–	–
	6082	B1	2/3	500–660	61	49	–	–	–	–	–	–	–	–	C2	5/10	600–670	106	54	169	6
	6083	B2	3/5	450–530	235	–31	14	34	–	–	–	–	–	–	C2	3/5	600–670	106	56	253	8
	average	B2	4*/16/43	–	232	–39	132	6	3*/8/15		219	34	32	14	C2R	3*/9/16	–	108	56	571	3
		B2	4/16*/43	–	232	–38	33	7	3/8*/15		217	35	13	16	C2R	3/9*/16	–	106	55	211	4
		B1	2/3*/5	–	60	53	–	–	–		–	–	–	–	C2N	2/5*/5	–	286	–52	31	14
	Total average	B2	7*/27/63	–	226	–37	64	7	7*/25/51		221	38	44	8	C2N,R	5/14*/21	–	106	54	80	5
		B2	7/27*/63	–	230	–38	27	6	7/25*/51		221	39	17	7	C2N,R	5/14*/21	_	106	54	80	5
		B1	3/7*/16	–	57	48	38	10	–	–	–	–	–	–	–	–	–	–	–	–	–
	Southern poles	Pole B1		42°S, 311°E, dp/dm = 9°/13°					Pole D		2°S, 351°E, dp/dm = 5°/9°				Pole C1		18°N, 55°E, dp/dm = 5°/7°				
		Pole B2		39°S, 327°E, dp/dm = 4°/7°											Pole C2		22°N, 87°E, dp/dm = 5°/6°				

Note: *M*, *N*, and *n* are the number of beds, pits, and samples, respectively; D° and I° are the declination and inclination of average directions of the J_n components, respectively; K is the vector concentration; α_{95}° is the confidence circle radius at a 95% probability for the average direction; Φ° and Λ° are the latitude and longitude of the paleomagnetic pole, respectively; dp° and dm° are the confidence oval semiaxes; the statistic is given at the hand specimen level; the statistic level for average directions for the components (beds or hand specimens) is marked by an asterisk; B1, B2, C1, C2, and D are the J_n components.

Table 2. Reversal and consistency tests for the J_n components of the Narva sedimentary rocks

Compared components	N_1/N_2	γ_{cr}, °	γ, °	Reversal test	Consistency test
B1/B2	7/27	11	13	negative	
B2(O)/B2(Є)	11/16	11	11		positive, class C
C1(O)/C2(Є)	24/14	6	21		negative
C2N/C2R	5/9	14	3	positive, class C	
D(O)/D(Є)	17/8	18	7		positive, class C

Note: N_1 and N_2 are the number of pits for the compared average directions of the J_n components; γ and γ_{cr} are the measured and critical angles [McFadden and McElhinny, 1990] between the axes for normal and reverse directions.

of the Narva section. In addition, the absence of a statistically significant difference between the average directions of the B component for the Ordovician and Cambrian sections (Table 2) considerably decreases the probability that the B component is of the Cambrian age.

EARLY PALEOZOIC APWP SEGMENT FOR BALTICA AND ITS KINEMATICS

The positions of the paleomagnetic poles obtained by us for the Baltic–Scandinavian province (22°N, 87°E (dp/dm = 5°/6°) for the Late Cambrian and 18°N, 55°E (dp/dm = 5°/7°) for the Tremadocian and Arenigian, see Table 1, Fig. 5) together with the recently published paleomagnetic poles for the same part of the province obtained in the Arenigian and Llanvirnian limestones [Smethurst et al., 1998; Lubnina et al., 2004; Gurevich et al., 2005] and the poles obtained using the rocks from the western Baltic–Scandinavian province (see [Torsvik and Rehnstrom, 2003] and references therein) determine the Late Cambrian–Early Silurian APWP segment for Baltica (Table 3). The shape of the more ancient part of APWP remains unclear since reliable results are still insufficient [Smethurst et al., 2001; Popov et al., 2002; 2005; Nawrocki et al., 2004; Iosifidi et al., 2005]. A recent analysis of the available Vendian paleomagnetic data for Baltica indicates that these data are contradictory [Walderhaug, 2004]. Therefore, it is evident that certain conclusions concerning the APW segment for Baltica at the end of the Proterozoic and at the very beginning of the Paleozoic should be based on the larger volume of high-quality data. One of the main problems related to the Baltic data, which was encountered in many works, consists in the selection of the time during which the J_n components were acquired. The prevalence of the rock remagnetization processes in the Baltic–Scandinavian province is also illustrated by the multicompositional analysis of rock sample magnetization presented in this work. The present and other works identify the remagnetization events with ages from the Late Cambrian to the Triassic (Table 4, Fig. 6). Similar complex magnetization histories were previously found in many other works [Bucha et al., 1976; Khramov, 1977; Smethurst et al., 1998]. However, the origin of these overlapped magnetizations is still disputable. It is assumed that remagnetization was sometimes related to tectonic events [Torsvik and Rehnstrom, 2003; Lewandowski and Abrahamsen, 2003]; however, this explanation is apparently inapplicable for our case since the region of sampling is located within the platform and outside the regions affected by the Paleozoic or Mesozoic active tectonic processes. Therefore, we supposedly relate remagnetization to the paleogeographic position of Baltica in hot and dry climatic belts in the Late Cambrian (our C2 component), Late Silurian (component D), and Late Paleozoic–Early Mesozoic (components B1 and B2). In this scenario remagnetization was caused by the oxidation of magnetic minerals. A similar process possibly resulted in remagnetization of the Vendian rocks (Table 5) and was discussed recently [Walderhaug, 2004].

The history of the Baltica's drift during the first half of the Paleozoic is characterized by the displacement of this plate from high and middle southern latitudes in the Early Paleozoic to tropical and equatorial latitudes in the Devonian (see, e.g. [Torsvik et al., 1996; Gurevich et al., 2005]). This drift's history was traditionally described as a combination of displacement (latitudinal motion) and rotation around the plate center (see, e.g. [Torsvik et al., 1996; Smethurst et al., 1998]). This description has a tectonic sense rather for small plates, blocks, overthrust plates, etc., which makes it possible to determine the initial affiliation of these formations, to trace their motions, and to estimate their deformations at the stage of collision. Gurevich et al. [2005] indicated that such a two-component representation for the motion of a large plate such as Baltica becomes meaningless, especially if this plate moved at high latitudes when "rotations" become artifacts (see, e.g. [Khramov, 1987]). As an alternative we can present the

Table 3. Paleomagnetic data for the Cambrian and Ordovician in the Baltic region

Object	Age, Ma	α_{95}°	Q	Pole coordinates		dp/dm	APWP		Reference
				Φ,°N	Λ, °E		Φ, °N	Λ, °E	
Limestones, Sweden, N	456	13	5	3	35	13/18	1	33	[Torsvik, Trench, 1991a]
Vestergotland N3, Sweden	459	5	6	5	34	5/7	5	37	[Torsvik, Trench, 1991b]
Vestergotland N1,2; R, Sweden	466	4	6	14	49	5/6	13	47.1	[Torsvik, Trench, 1991b]
Limestones, Leningrad region	468	6	5	9	59	7/10	16	49	[Lubnina et al., 2004]
Limestones Komstad, Sweden	471	9	6	19	51	11/15	20	50	[Torsvik, Rehnström, 2003]
Limestones Gullhogen, Sweden	472	7	6	19	54	8/10	22	51	[Torsvik et al., 1995]
Limestones I(R), Sweden	473	5	6	18	46	6/8	24	51	[Torsvik, Trench, 1991a]
Limestones, Putilovo, Russia	474	3	5	25	58	4/5	25	52	[Gurevich et al., 2005]
Limestones, Sweden	475	2	5	30	46	3/4	27	52	[Claesson, 1978]
Limestones, Sweden	476	9	5	30	55	12/15	28	53	[Perroud et al., 1992]
Limestones, Narva, Russia	477	4	6	18	55	5/7	29	54	this work
Limestones, St. Petersburg, Russia	478	4	6	35	59	5/6	30	56	[Smethurst et al., 1998]
Seiland complex C, Norway	490	11	5	31	86	11/15	32	76	[Torsvik et al., 1990]
Sandstones, Narva, Russia	500	5	5	22	87	5/9	23	88	this work

Note: The age (Ma) is given according to Tucker and Mc Kerrow [1995]; α_{95} is the confidence circle radius at a probability of 95% for the average direction; Q is the sum of the Van der Voo reability criteria [Van der Voo, 1990]; Φ° and Λ° are the latitude and longitude of the southern paleomagnetic pole, respectively; Φ_p° and Λ_p° are the latitude and longitude of the southern paleomagnetic pole for APWP, obtained using the spherical spline method [Jupp and Kent, 1987].

motion of Baltica in the Early Paleozoic as a clockwise turn by 68° around the Euler pole with coordinates 43°S and 165°E (the longitude of the Euler pole corresponds to the position of St. Petersburg 456 Ma ago at the present-day longitude). A large drift amplitude (Fig. 7) supports the hypothesis of "the Inertial True pole wandering" (IITPW [Kirschvink et al., 1997]) during the Early Paleozoic. However, our model is an extreme one among other possible models; thus, this model cannot, nevertheless, confirm the IITPW hypothesis. Meert [1999] discussed the assumption that rapid reorganization of the location of the lithospheric plates between

Table 4. South paleomagnetic poles for the secondary J_n components of the Ordovician rocks for Baltica

No.	Object	Component, polarity	Rock age, Ma	Magnetic age, Ma	Pole coordinates		dp	dm	Reference
					Φ,°S	Λ, °E			
1	Limestones, Sweden	III, N	464–485	180	68	325	9	12	[Torsvik, Trench, 1991a]
2	Limestones, St. Petersburg	S, N	470	180	63	345	2	3	[Smethurst et al., 1998]
3	Limestones, Putilovo	S, N	475	200	60	330	3	4	[Gurevich et al., 2005]
4	Limestones, Putilovo	S, N	465	220	58	344	2	2	[Gurevich et al., 2005]
5	Slates, Podolia	S, N	485	220	57	348	3	4	[Gurevich et al., 2005]
6	Narva	BI, N	480–500	220	42	311	9	13	this work
7	Limestones, St. Petersburg	S, R	470	240	58	331	4	5	[Smethurst et al., 1998]
8	Limestones, Bornholm	1B, R	471	255	46	329	4	8	[Torsvik, Rehnström, 2003]
9	Narva	B2, R	480–500	260	39	327	4	7	this work
10	Limestones, Bornholm	IBV, R	475	275	40	343	3	5	[Lewandowski, Abrahamsen, 2003]
11	Vestergotland	IB, R	460–476	287	35	351	2	4	[Torsvik, Trench, 1991b]
12	Limestones, Sweden	II, R	464–485	290	34	348	10	19	[Torsvik, Trench, 1991a]
13	Limestones, Bornholm	IBV, R	475	350	28	339	3	5	[Lewandowski, Abrahamsen, 2003]
14	Limestones, Andrarum	I B,R	500	400	17	342	3	5	[Torsvik, Rehnström, 2001]
15	Seiland complex, Ç	C, R	486–612	410	5	335	5	9	[Torsvik et al., 1990]
16	Narva rocks	D, R	480–500	420	2	351	5	9	this work
17	Limestones, Skania	1B, R	471	450	4	12	4	6	[Torsvik, Rehnström, 2003]
18	Limestones, Bornholm	HBV, R	475	460	–11	32	7	10	[Lewandowski, Abrahamsen, 2003]
	Average 1–6	N	–	180–220	59	333	9		
	Average 7–12	R	–	240–290	42	339	8		

Table 5. The Cambrian and Late Vendian (500–600 Ma) paleomagnetic poles selected for Baltica

Object	Index in Fig. 6	α°_{95}	Q	Pole coordinates		dp/dm	Reference
				Φ, °N	Λ, °E		
Alum Slates	*1*	7	3	52	111	7/10	[Torsvik et al., 2001]
Tornetrask Formation	*2*	9	4	56	116	12/15	[Torsvik et al., 2001]
Nyborg Formation	*3*	18	5	24	89	17/25	[Torsvik et al., 1995]
Stappogiedde Formation, exp. 6	*4*	11	3	–1	78	6/11	[Bylund, 1994]
Alno complex	*5*	9	4	–7	92	5/9	[Piper, 1981]
Stappogiedde Formation, exp. 7, 9	*6*	10	4	–26	129	8/12	[Bylund, 1994]
Winter Coast, sedimentary rocks, component Z	*7*	3	6	–25.3	132.2	2/4	[Popov et al., 2002]
Zolotitsa, redbeds, component Z	*8*	2	6	–32	113	2/3	[Popov et al., 2005]
Podolia, sedimentary rocks, component P2	*9*		5	–40	97	$A_{95} = 8$	[Iosifidi et al., 2005]
Volynian basalts, A	*10*	2	4	–36	153	3/4	[Nawrocki et al., 2004]
Volynian basalts, A	*11*	4	4	–27	102	2/4	[Nawrocki et al., 2004]

Note: See Table 3.

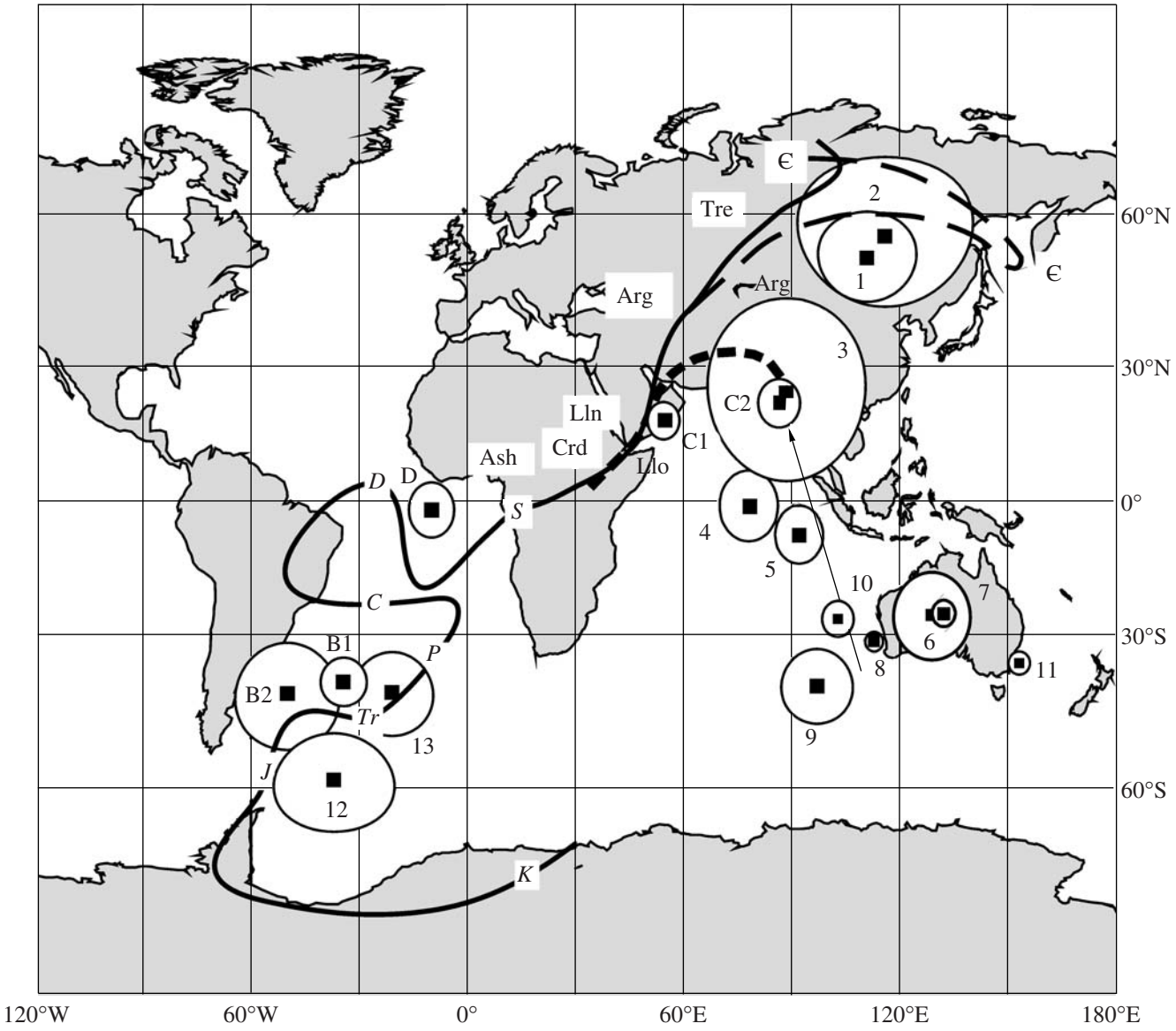


Fig. 6. APWP [Smethurst et al., 1998] and new poles for Baltica. Dotted lines represent the classical curve, including the data on the Fen Formation (upper part), and the proposed alternative APWP version from the Late Cambrian to the Early Silurian (data from Table 4, lower part); B1, B2, C1, C2, and D are new paleomagnetic poles according to the Narva rocks (Table 1); (1)–(11) the Late Vendian and Cambrian poles for Baltica (Table 4); (12) and (13) the average paleomagnetic poles for the secondary J_n components (Table 5); an arrow shows the possible APWP version in the interval the Late Vendian–Cambrian for Baltica.

523 and 506 Ma ago was caused by an IITPW episode. However, the review of the paleomagnetic data for the Laurentia, Baltica, Siberia, and Gondwana plates [Meert, 1999] indicated that none of the APW segments for these plates reaches the necessary length, thereby disproving IITPW for that time interval. Unfortunately, only one paleomagnetic determination exists in the interval 523–506 Ma for Baltica [Torsvik and Rehnstrom, 2001], and at least two different versions of the Vendian–Cambrian APWP segment were proposed for Baltica (Fig. 6). These versions can be completed with two more versions obtained by changing polarities of the Winter Coast poles and the poles of the same age [Nawrocki et al., 2004; Popov et al., 2005].

Several years ago, the dualism of the Vendian–Cambrian paleomagnetic data was revealed not only in Baltica (Table 5) but also in Siberia and Laurentia [Meert et al., 1994; Pisarevsky et al., 1998; 2000]. Pavlov et al. [2004] assumed that both fields, represented by two alternative APW paths, are Late Vendian–Early Cambrian fields: they replaced each other many times during this period. However, this brave hypothesis only shifts the problem from the field of rock magnetism to the theories of geomagnetism and geodynamics: i.e., which of the above fields was the axial dipole field, and

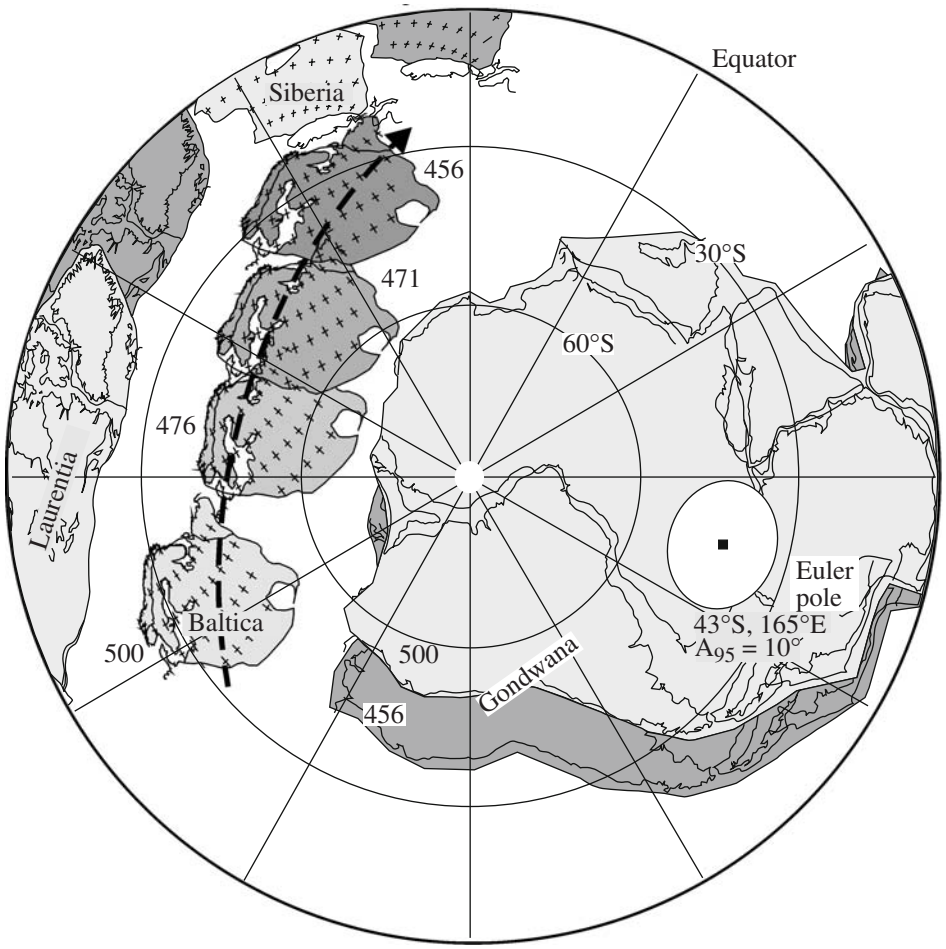


Fig. 7. The possible version of the Baltica drift in the Early Paleozoic. The model of the remote Euler rotation pole. Longitudes are arbitrary; they (including the Euler pole longitude) correspond to the St. Petersburg position 456 Ma ago at the present-day longitude. The positions of Gondwana, Laurentia, and Siberia for the beginning (500 Ma ago) and end (456 Ma ago) were taken from [Khramov, 2007].

why another nonaxial/nondipole field could repeatedly originate and disappear, having an unstable configuration during such relatively prolonged periods.

CONCLUSIONS

The paleomagnetic studies of the Cambrian and Lower Ordovician rocks in the exposures on the right bank of the Narva River together with the recent paleomagnetic data on the Lower Paleozoic for other regions of the Baltic–Scandinavian province made it possible to specify the path of apparent migration of the Baltic paleomagnetic pole for the Early Paleozoic and to propose a new version of the path segment for the Late–Early Ordovician. This segment is of special significance for the paleogeographic reconstruction of the ensemble of the Baltica, Laurentia, and Gondwana lithospheric plates at the beginning of the Paleozoic and for the history of the interrelation between these plates and the elements of Avalonia and Armorica during the Paleozoic. The earlier pre-Late Cambrian segment of APWP for Baltica remains conditional, and the shape of this segment can be obtained based on new high quality data. We represent the history of the Baltica's drift during the Late Paleozoic as a clockwise turn by 68° around the Euler pole with coordinates 43°S and 165°E. This model is at the boundary of the possible versions and, consequently, cannot nevertheless confirm the IITPW hypothesis. We relate the remagnetization identified in the studied samples to the secondary processes of oxidation during the Silurian–Devonian and Permian–Triassic, when the studied region was located at latitudes for which hot and dry climatic conditions were typical.

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